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Exercise prescription and public health

Learning outcomes

By the end of this session, you will be able to:

- Define exercise prescription and explain its relevance to public health.
- Summarize key physical activity guidelines for adults and special populations.
- Apply exercise prescription principles (FITT-VP) within a public health framework.
- Discuss physiotherapists' contributions to community and national health promotion initiatives.

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Why it matters to public health

- 1 in 4 adults globally do not meet recommended physical activity levels (WHO, 2023).
- Physical inactivity contributes to ~5 million deaths annually.
- Rising chronic disease burden (CVD, diabetes, MSK pain)
- Shift from treatment to prevention (NHS long term plan)

Mentimeter

What public health issues do you associate with inactivity

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Exercise prescription

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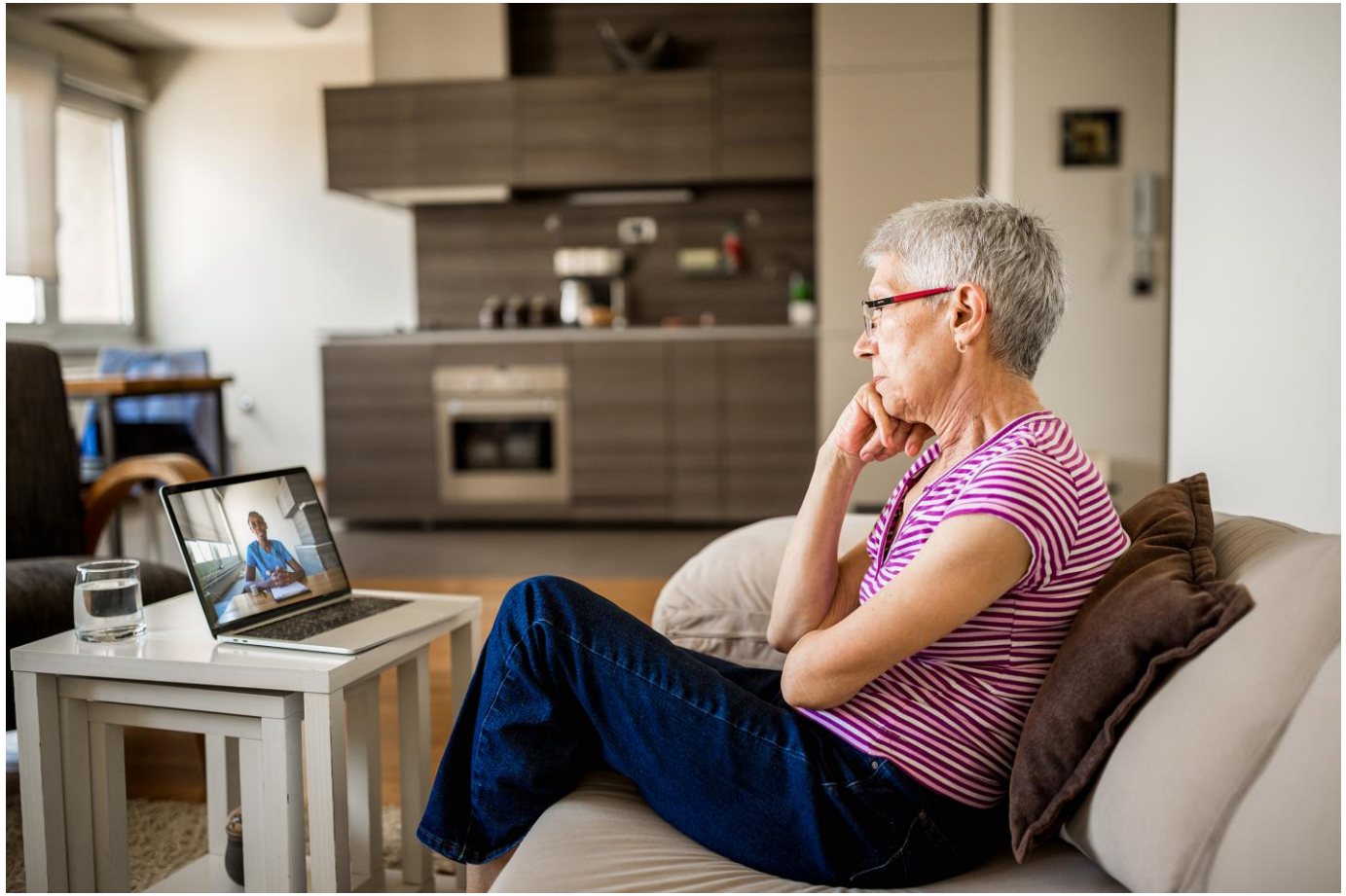


What is exercise prescription ?

- Definition: “Systematic, individualized guidance on physical activity to achieve specific health or fitness outcomes.”
- Components: Type, intensity, frequency, duration, progression.
- Aim: Promote safe, effective, and sustainable activity.

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A practical example… meet Tracy



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Public health Exercise guidelines (WHO)

Some is good, more is better

Per week:

- 150-300 mins moderate-intensity

OR

- 75-150 minutes of vigorous intensity or a mix
- Muscle strengthening on at least 2 days per week
- Break up sedentary activity



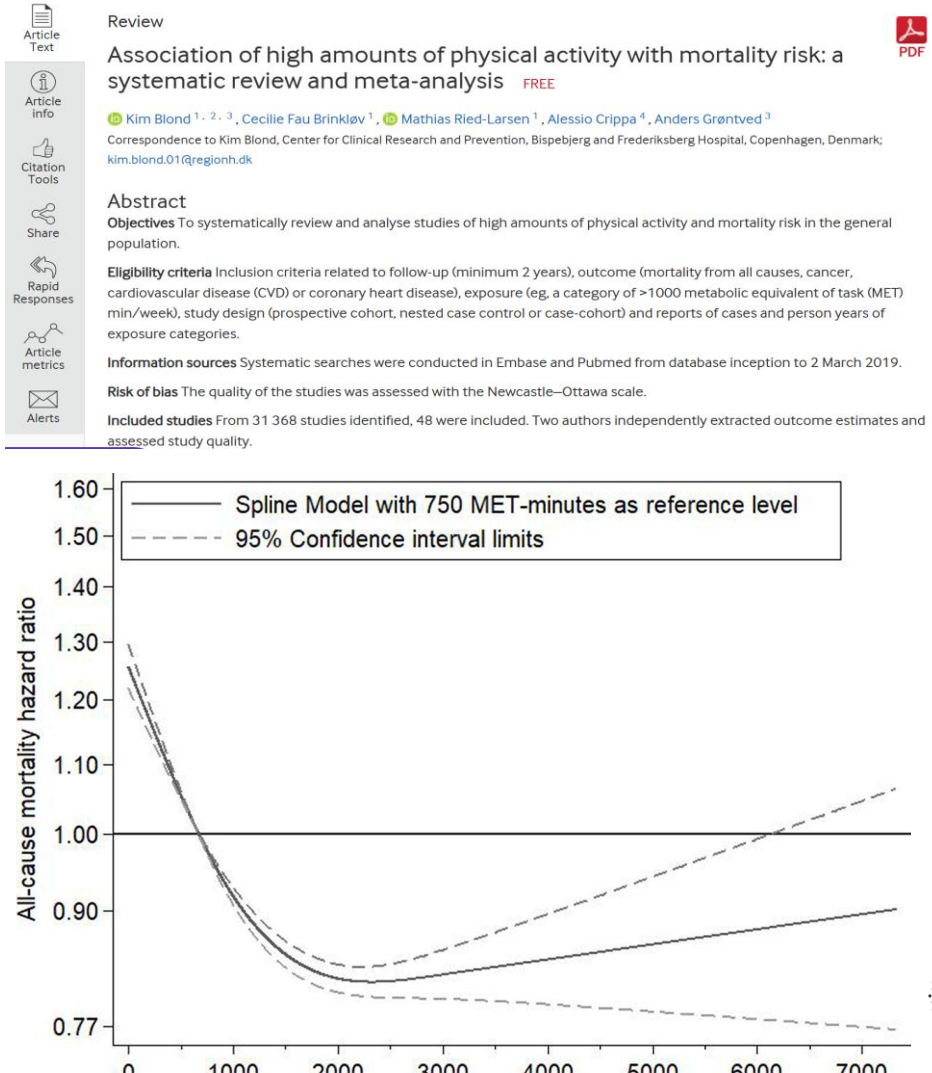
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Guidance for all ages

Group	Main target	Extra emphasis
Adults	150–300 min moderate or 75–150 min vigorous per week	Muscle strengthening 2+ days/week, reduce sitting bjsm.bmj+1
Children and young people	60 min/day on average	Aerobic activity, muscle and bone strengthening bjsm.bmj
Older adults	Same general adult target when possible	Balance, functional activity, strength work pmc.ncbi.nlm.nih+1

Some is good more is better

- The first steps away from inactivity have the biggest gains
- Gains increase with increased adherence
- Not aiming for perfection just participation



Review

Association of high amounts of physical activity with mortality risk: a systematic review and meta-analysis FREE

Kim Blond [1, 2, 3], Cecilie Fau Brinkløv [1], Mathias Ried-Larsen [1], Alessio Crippa [4], Anders Grøntved [3]
Correspondence to Kim Blond, Center for Clinical Research and Prevention, Bispebjerg and Frederiksberg Hospital, Copenhagen, Denmark; kim.blond.01@regionh.dk

Abstract

Objectives To systematically review and analyse studies of high amounts of physical activity and mortality risk in the general population.

Eligibility criteria Inclusion criteria related to follow-up (minimum 2 years), outcome (mortality from all causes, cancer, cardiovascular disease (CVD) or coronary heart disease), exposure (eg, a category of >1000 metabolic equivalent of task (MET) min/week), study design (prospective cohort, nested case control or case-cohort) and reports of cases and person years of exposure categories.

Information sources Systematic searches were conducted in Embase and Pubmed from database inception to 2 March 2019.

Risk of bias The quality of the studies was assessed with the Newcastle–Ottawa scale.

Included studies From 31 368 studies identified, 48 were included. Two authors independently extracted outcome estimates and assessed study quality.

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Exercise prescription

How can Physiotherapists influence activity levels in our sessions ?

Responses can be up to 200 characters and will appear here.

You can group responses if you get more than 10.

Turn on voting so people can flag their favorite responses.

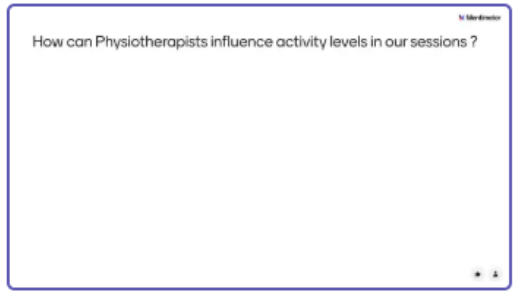
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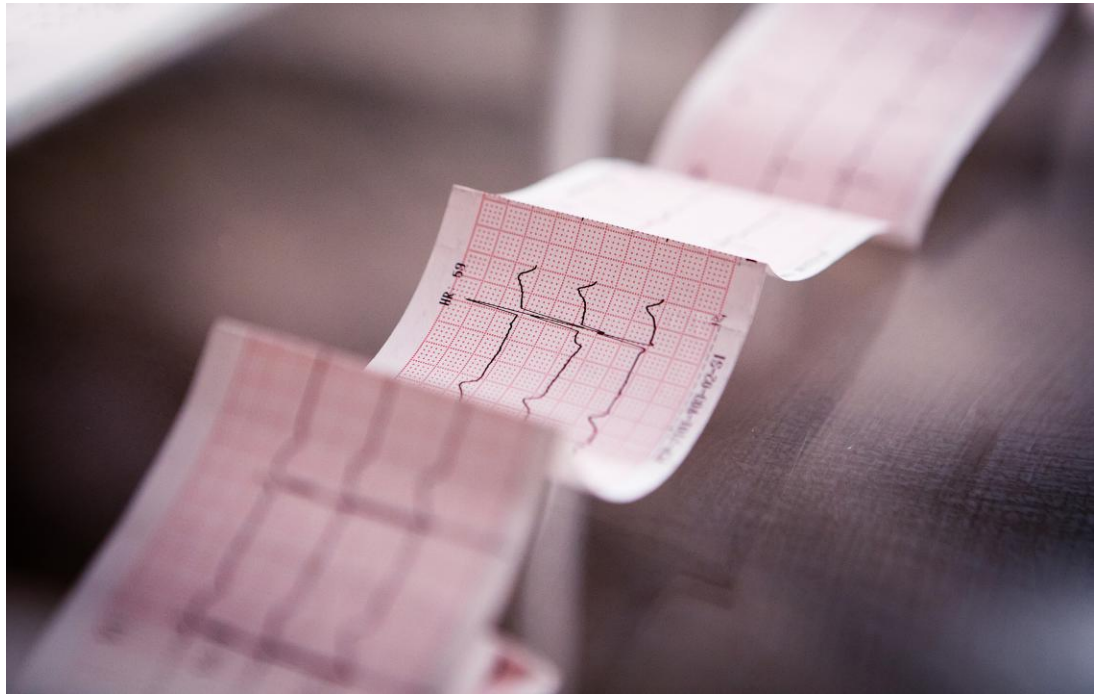
Why Exercise matters to public health

- Physical inactivity is a major modifiable risk factor for non-communicable disease.
- It contributes to cardiovascular disease, type 2 diabetes, obesity, some cancers, poor mental health, and reduced functional capacity.
- Public health approaches aim to shift whole populations, not just treat individuals.

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Exercise as prevention: CVD and Metabolic Health

- Lowers risk of heart disease and stroke by 19%.
- Lowers risk of type 2 diabetes by 17%.
- Maintain a healthy body weight
- Improves measures of body fat
- Physical inactivity is a leading risk factor for non-communicable disease and death worldwide



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Exercise and general health



- Regular physical activity reduces risk of many cancers by 8–28%.
- It contributes to prevention and management of cardiovascular disease, cancer, and diabetes.
- Public health evidence also links activity with better metabolic health and reduced premature mortality.

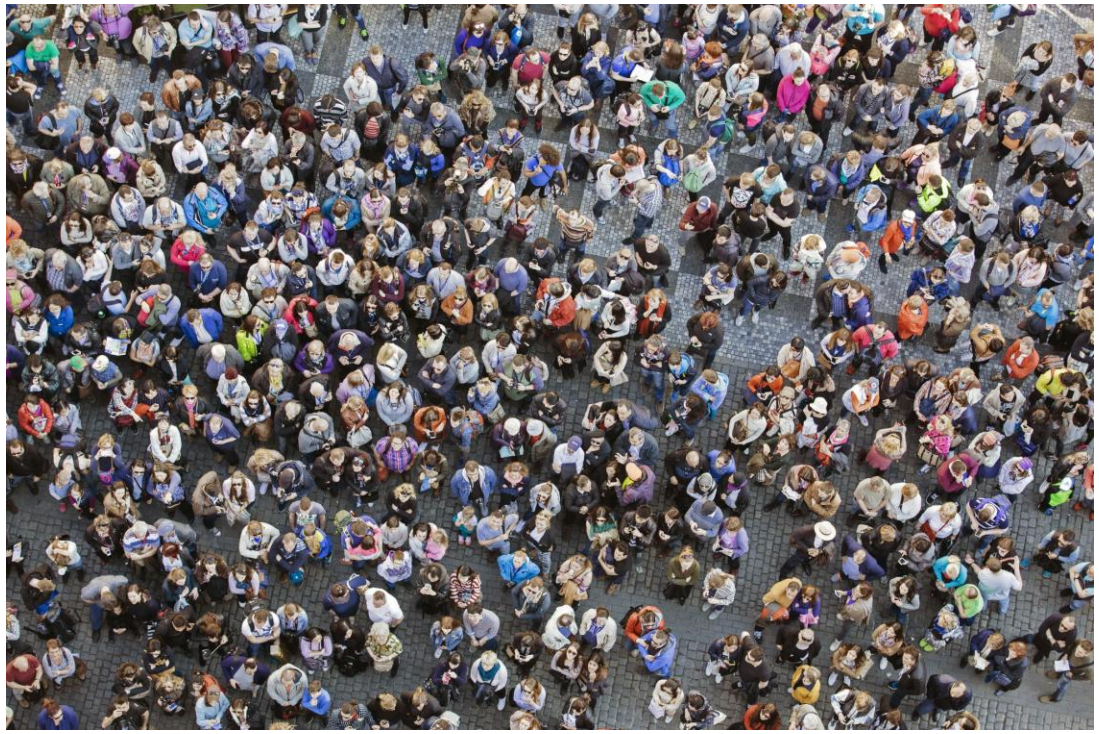
Exercise as prevention: Mental health

- Regular physical activity reduces risk of depression and dementia by 28–32%.
- It improves mental health, cognitive health, sleep, and overall well-being.
- In older adults, physical activity is associated with fewer falls and better mobility, balance, independence, and quality of life.



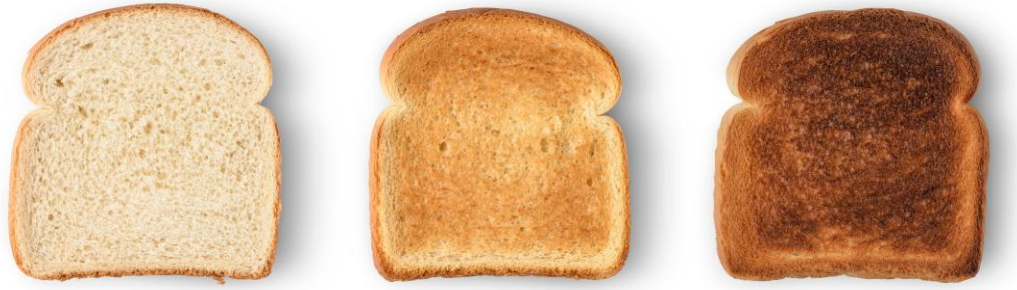
From individual care to population impact

- In the clinic, physiotherapists work with one person at a time.
- In public health, the aim is to improve health outcomes across groups and communities.
- Small changes across many people can produce a large impact.



The public health model: 3 levels

- Primary prevention: stop problems developing.
- Secondary prevention: detect and act early.
- Tertiary prevention: reduce complications and disability.



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Physiotherapy role

- Educator.
- Motivator.
- Behaviour-change supporter.
- Advocate for accessible activity.
- Contributor to community and health service initiatives.



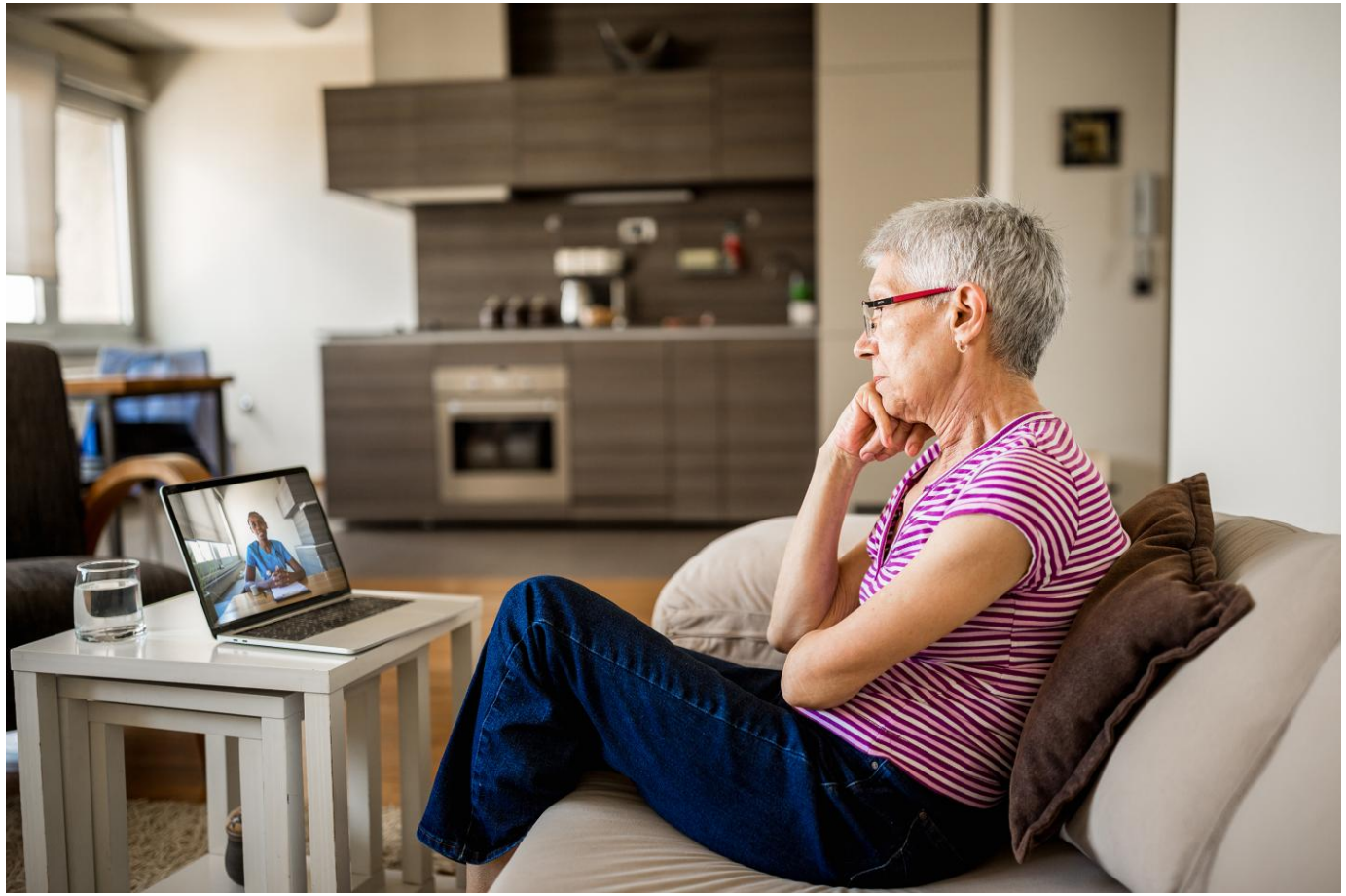
Brief intervention framework

- Ask.
- Advise.
- Act.
- Support.



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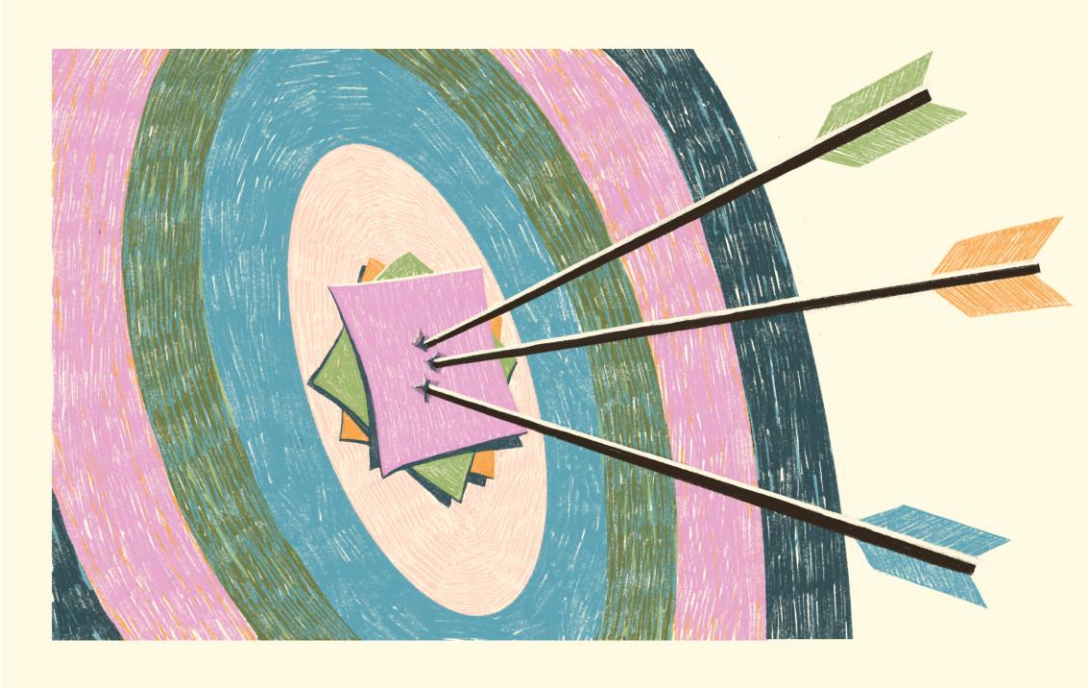
Let’s talk about Tracy



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The How of Exercise prescription

- Exercise prescription is a structured, individualised plan.
- It should be based on goals, baseline activity, health status, and risk.
- In public health, the aim is not only performance but also safety, adherence, and long-term behaviour change



The FITTVP framework



Frequency: how often.



Intensity: how hard.



Time: how long.



Type: what kind of activity.



Volume: total amount.



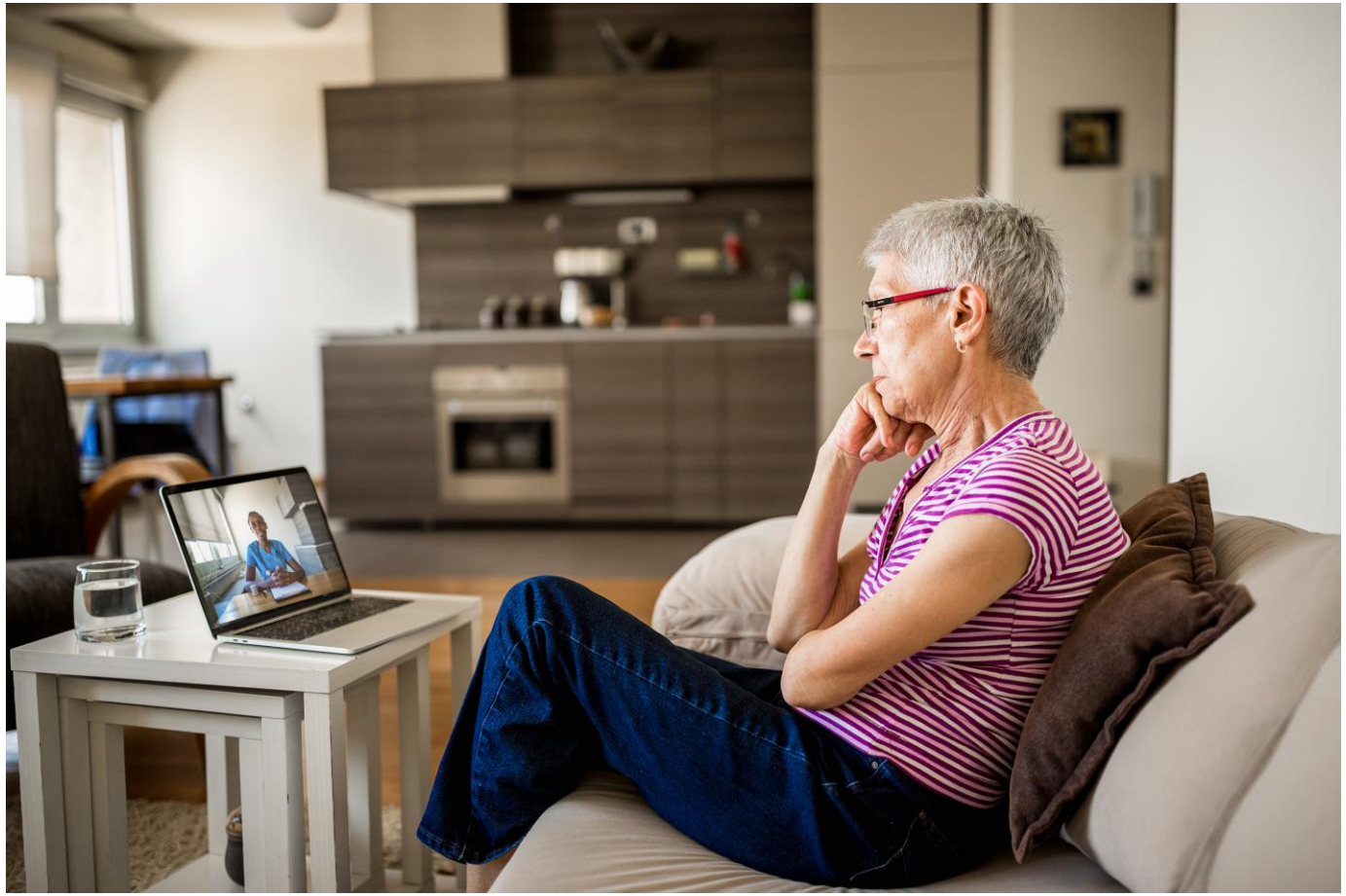
Progression: how it changes over time.

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Applying FITT-VP in practice

- Start with what the person can realistically do.
- Use public health guidelines to inform targets, but tailor the starting point.
- Progress gradually to improve adherence and reduce dropout
- Use SMART goals

Let’s talk about Tracy



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Why tailor exercise prescription

- Public health guidance applies across the lifespan, but needs adapting.
- Different groups face different barriers, risks, and priorities.
- Good prescription balances safety, relevance, equity, and adherence.

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Children and young adults

- Children and adolescents should average **60 minutes per day** of moderate-to-vigorous physical activity across the week.
- Most activity should be aerobic, with vigorous and muscle- and bone-strengthening activities included at least 3 days per week.
- Active play, PE, sport, walking, and cycling all count.

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Older adults

- Older adults should aim for the same aerobic targets as other adults when able: **150–300 minutes moderate** or **75–150 minutes vigorous** activity per week.
- They should also do strength work and include **balance and functional activity**, particularly if at risk of falls.
- Sedentary time should be reduced and broken up regularly.

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Long term conditions and disability

- People living with chronic conditions or disability should still be as active as their abilities and conditions allow.
- WHO recommends the same broad adult targets where possible, adapted to symptoms, function, and risk.
- For disabled adults, UK evidence reviews support **150 minutes of moderate-to-vigorous activity**, plus **2 sessions per week of challenging strength and balance exercise**, where appropriate.

In a nutshell

Group	Main emphasis	Example message
Children and young people	Daily movement, play, development	“Make movement fun and routine.” bjsm.bmj+1
Older adults	Strength, balance, function, falls prevention	“Move often, stay strong, stay independent.” gov+1
Long-term conditions/disability	Adaptation, symptom management, access	“Modify activity, don’t avoid it.” gov+1

Group task

- In groups, tailor a brief exercise prescription for one of these populations:
- a 14-year-old with low activity levels,
- a 78-year-old with falls risk,
- a 55-year-old with type 2 diabetes and obesity.

- Identify :
  - the main public health goal,
  - one key barrier,
  - one appropriate starting activity,
  - and one progression idea

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Take home messages

- As physios we are perfectly placed to provide PH advice and guidance especially around physical activity
- It's part of our HCPC SOPs to provide health advice and guidance

- Treat the individual not the issue
- A little bit more movement is better than nothing at all

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Questions?

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